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LETTER

Combining animal movements and behavioural data to detect behavioural states

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Abstract

Animal movement paths show variation in space caused by qualitative shifts in behaviours. I present a method that (1) uses both movement path data and ancillary sensor data to detect natural breakpoints in animal behaviour and (2) groups these segments into different behavioural states. The method can also combine analyses of different path segments or paths from different individuals. It does not assume any underlying movement mechanism. I give an example with simulated data. I also show the effects of random variation, # of states and # of segments on this method. I present a case study of a fisher movement path spanning 8 days, which shows four distinct behavioural states divided into 28 path segments when only turning angles and speed were considered. When accelerometer data were added, the analysis shows seven distinct behavioural states divided into 41 path segments.

Keywords

Animal movement, behavioural states, breakpoints, correlated random walk, segments, spatial scale, turning angles.

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INTRODUCTION

Animal movement is the glue that binds together both behaviour at a lower scale of organisation and population dynamics at a higher scale of organisation. One key aspect of animal movement is variation throughout space caused by different behavioural states. Sometimes these states are obvious. For example elk (*Cervus elaphus*) switch between exploration, foraging and social bonding behaviours (Fryxell *et al*. 2008). But more often the underlying randomness in movement patterns and habitat make behavioural states and the transitions between them difficult to detect by eye. Thus, behavioural ecologists require statistical tools to detect the underlying behavioural states that cause variation in animal movement.

There are two parts to this task: (1) dividing a time period into segments and (2) classifying those segments into discrete behavioural states. Behavioural ecologists have used many different variables to do this, and the variables depend on the spatial scale. At small spatial scales, studies typically use direct observations of behaviour to segment behaviours. For example, Schaefer & Messier (1995) measured rates of pawing to classify muskoxen feeding behaviours. However, at larger spatial scales it is difficult to watch animals, so studies typically use location data from telemetry to segment behaviours. Examples of variables estimated from movement data that have been used in behavioural state analyses include turning angles and step lengths (Frair *et al*. 2005), speed and direction vectors (Thiebault & Tremblay 2013), the time spent in the vicinity of successive path locations (Barraquand & Benhamou 2008) and fractal dimension (Tremblay *et al*. 2007).

Recent technological advances in sensors have made it possible to gather more than just location data from telemetry devices. Such sensors include activity (Adrados *et al*. 2003), acceleration in different directions (Löttker *et al*. 2009), pulse, respiration and body temperature (Nagl *et al*. 2003) and diving depth (Witteveen *et al*. 2008). No current statistical method combines both locational and behavioural data.

Current techniques for segmenting location data differ in how much is assumed about the underlying movement mechanisms, whether the numbers of states are assumed or estimated, and whether partitions between the states are estimated. One such technique assumes that the underlying mechanism of movements for the different behaviour states are known, and then directly estimates parameters for them using a mixture of movement models in a Bayesian framework. This has typically been done using correlated random walks (CRW) as the basic movement model for each behavioural state, although the method may also be applied to other movement models. CRWs are movements where an animal makes discrete steps, and at each step, the turning angle is independent of the previous turning angle. As well, successive step lengths are not auto-correlated or cross-correlated with turning angles or moving directions (Kareiva & Shigesada 1983).

For example, Morales *et al*. (2004) fitted CRWs to elk movement data using two behavioural states – an 'encamped' state with short step lengths and high turning angles, and a travelling state with long step lengths and low turning angles. These techniques have even been applied to humans, with Vermard *et al*. (2010) modelling fishing vessels using three behavioural states – fishing, steaming and stopping. The strengths of the mixed movement models are that, because they make strong assumptions about the movement mechanisms, they are statistically powerful to estimate movement parameters and boundaries between behavioural states. In addition, one can find how the parameters of the movement models relate to habitat variables. A key limitation is the often prohibitive computation required. Also, these methods may give erroneous results when applied to paths governed by other movement models. There is no test to see if alternate mechanisms better fit the data. For example, many animals travel with oriented paths (Nams 2006), which are modelled differently from CRWs.

Some techniques have been developed to detect breakpoints in movement paths without requiring assumptions about the movement mechanisms. For example, Barraquand & Benhamou (2008) applied techniques developed to process signals (Lavielle 2005) to detect breakpoints in movement paths. Gurarie *et al.* (2009) developed a likelihood-based technique to detect breakpoints. The strength of these techniques is that they are not limited to specific movement mechanisms. The limitation is that these techniques do not group the resulting segments into behavioural states.

I present a general statistical method that uses both location and other ancillary sensor data to objectively detect natural breakpoints in animal movement paths, and then to group these segments into different behavioural states. The method combines a version of direct search optimisation (Kolda *et al.* 2003), iterative sampling and constrained k-means clustering. This method does not assume any underlying movement mechanism. Thus, one can classify the different behaviour states and then later figure out their movement mechanisms. The method can also combine analyses of different path segments or paths from different individuals, thus, defining behavioural states similarly across the different path segments. Although I developed this method in order to analyse animal movement paths, the method can use any behavioural data and so can also be used to analyse behavioural states for non-movement behavioural studies.

THE METHOD

The basic conceptual model is of an animal with a certain number of qualitatively different movement patterns (behavioural states) that switch from one to another. For each behavioural state, there is a mean value for each variable. The parameters to be estimated are the behavioural state means and the positions of the switches in behavioural states – i.e. the partition points between the movement segments. The procedure finds the most parsimonious set of these parameters, with parsimony being measured by the Bayesian information criteria (BIC; Burnham & Anderson 2002). The procedure does this by testing a series of # of states and # of segments. For each combination of # of states and # of segments, an iterative procedure finds the most parsimonious set of partitions. Then, the most parsimonious combination of the # of states and # of segments is selected. I will call this method the Behavioural Movement Segmentation method – BMS (contact the author for a copy of the Mathematical code). All analyses were done using Mathematica 9.01 (Wolfram Research 2012).

The simulation data are a series of values for variables measured at equally spaced times. For example, if the variables are turn angle, step length and level of activity, then the data are a series of turning angles, step lengths and activity levels, one measured at each time. The data are normalised to ensure valid BIC estimates. Then the data are standardised to zero mean and unit variance for each variable. This gives all variables equal weighting in the analysis.

Although this is a numerical nonlinear estimation method, it cannot be solved easily using traditional numerical methods because the numbers of parameters to be estimated are also unknown – they depend on the # of states. Thus, this method travels through the solution space finding the combinations of parameters that minimise BIC.

Definitions

n = # of locations in the movement path;
n_{smin} = within one n_b, the # of segments at which BIC is minimised;
n_{bmin} = the # of behavioural states at which BIC is minimised;
r = resolution of sampling – i.e. initially sample every r^{th} point along the path;
n_s = # of segments;
n_b = # of behavioural states;
L_{min} = minimum segment length.

The specific procedure

The procedure works hierarchically; thus, I will explain the different levels.

Finding the optimum number of behavioural states

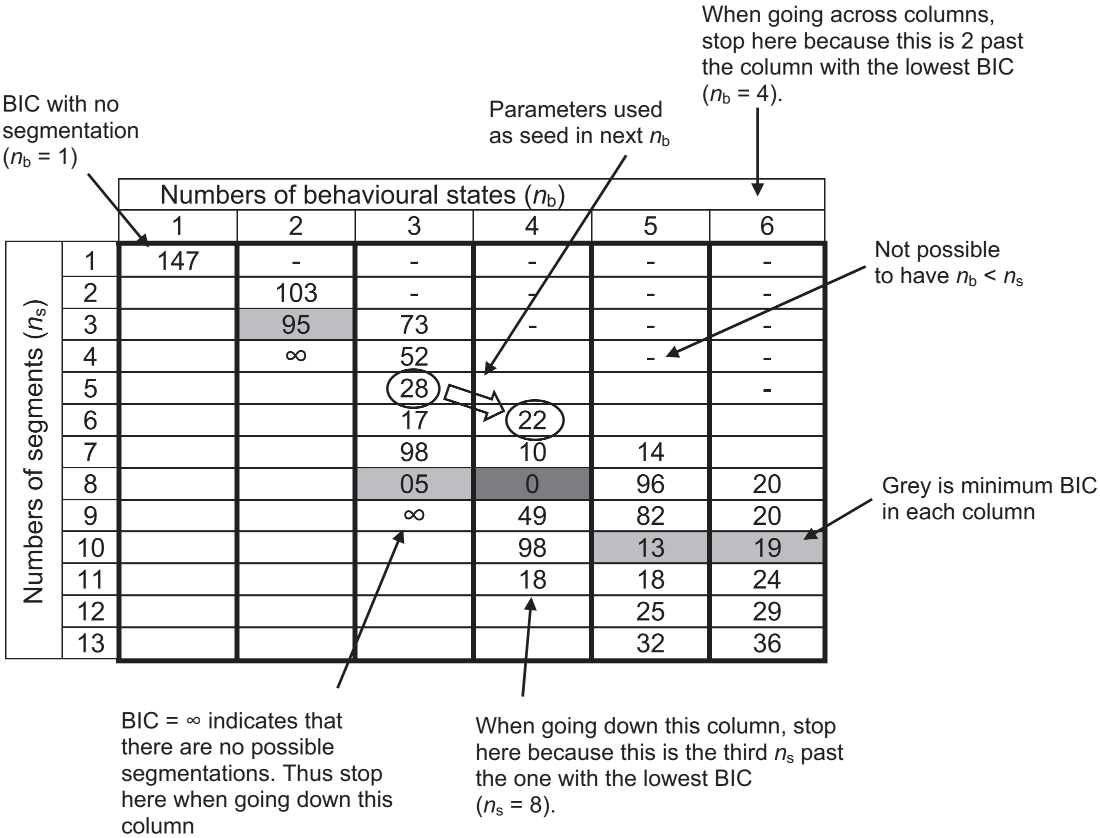
The procedure first estimates the BIC for one state and one segment. Then it starts with two behavioural states (i.e. $n_b = 2$) and finds the optimal number of segments and the specific partitions between the segments. This is repeated successively for higher numbers of behavioural states, and finally stops when $n_b - n_{bmin} = 2$. Thus the procedure works from larger to smaller scale – i.e. from fewer to more behavioural states (from the left columns to the right columns in Fig. 1).

Finding the optimum number of segments

For each n_b the optimum number of segments are found as follows. The procedure works from smaller to larger n_s values, finding the optimal segmentation for each n_s (i.e. going down each column in Fig. 1). The starting n_s and its corresponding optimum segmentation are taken from n_b-1. This initial n_s is the highest of: (1) n_{smin}-3 from n_b-1, or (2) the lowest possible n_s in n_b. The ending n_s occurs when: (1) n_s-n_{smin} = 3 or (2) an n_s is reached for which there is no possible segmentation (i.e. when all segmentations have some adjacent behavioural states that are the same).

Finding the optimal segmentation

For each n_s and n_b value, the optimal segmentation is found as follows. The procedure starts with the optimal segmentation for n_s-1. Then:
(1) it adds a new partition, selecting the best one (i.e. minimum BIC) from all available partition points.
(2) it replaces each partition with the best one chosen from the range of the two adjacent partitions. This is repeated until the segmentation does not change.
(3) it replaces each partition point with the best one chosen from all available partition points. This is repeated until the segmentation does not change.



n_s \ n_b	1	2	3	4	5	6
1	147	-	-	-	-	-
2		103	-	-	-	-
3		95	73	-	-	-
4		∞	52		-	-
5			28			-
6			17	22		
7			98	10	14	
8			05	0	96	20
9			∞	49	82	20
10				98	13	19
11				18	18	24
12					25	29
13					32	36

Figure 1 The Δ Bayesian information criteria (BIC) estimates from the example simulation analysis. ΔBIC is the difference between BIC and the minimum BIC. The method traverses each column, going from left to right. $n_b = n_s = 1$ represents no segmentation. In each column, the method goes from top to bottom. The starting and ending position in each column is determined by either the position of the minimum in the previous column. The method finishes when it has gone 2 columns past the minimum.

Finding the best available partition point

For one specific segmentation and n_s and n_b value and a specific range of possible points, the best partition point is found. For example, in (2) above, the range is from the previous to the next partition. Available partition points are all those within the range such that no segment is shorter than the minimum segment length. This is the step that takes a long time, so to minimise computing time this is done hierarchically. First the best point is found from every r^{th} point. Then, the best point is found from every point, but within the range of the previous and next r^{th} point. The resolution of sampling, r, is preset to the floor of 1/3 of the minimum segment length to ensure that there are at least three sampling points in each segment. However, r could be overridden by the user. Decreasing r might ensure that no segments are missed, but if r is too small, then the procedure will take too long.

Combining segments into behavioural states

For a given segmentation:
(1) the means for each variable for each segment are estimated.
(2) then these means are grouped into n_b clusters using k-means clustering (Mathematica function FindCluster; (Morissette & Chartier 2013), with the Manhattan Distance used for the similarity measure (Wang *et al.* 2007). The Manhattan Distance is the distance between two points measured along axes at right angles. It was chosen after testing seven similarity measures on simulated data. This step links the segmentation to the behavioural states.
(3) if any segments in the same state are adjacent to each other, then this segmentation is rejected. This step also links the segmentation to the behavioural states.
(4) the means for each state are estimated by grouping all the segments of each state together.
(5) the expected values for each location are estimated by using the means of each state.

Estimating BIC

The means for each variable for each segment are the expected model for each data point. The BIC criterion is:

$$\mathrm{BIC} = -2\log(\pounds(\mathrm{M})) + k\ \log(n) \tag{1}$$

where:

$\mathscr{L}(\mathrm{M})$ = the Likelihood of the predicted model
k = number of estimated parameters.

For univariate normal distributions eqn (1) simplifies to:

$$\mathrm{BIC} = \mathrm{n}\,\log(\sigma^2) + \mathrm{k}\,\log(\mathrm{n}) \quad (2)$$

however, unlike most other types of model selection, I have multiple response variables – i.e. predictions are estimated for each input variable. Thus, for multivariate normal distributions eqn (1) simplifies to:

$$\mathrm{BIC} = \mathrm{n}\,\det(\mathrm{cov}(\mathrm{O} - \mathrm{E})) + \mathrm{k}\,\log(\mathrm{n}) \quad (3)$$

where:
O = the observed data – i.e. the n × v matrix of input data
E = the expected data – i.e. the n × v matrix of estimated values
$k = (n_S - 1) + v\,n_S$
cov = sample covariance matrix.

The estimated parameters are the locations of the $n_s - 1$ partitions, and the n_s mean values of the states

Model selection

The BMS algorithm uses BIC as an estimate of model fit rather than the commonly-used small-sample Aikaikie Information Criteria (AICc). I use BIC because I found that the AICc overestimated the # of behavioural states and # of segments. Also, and most importantly, AICc detected segments when the simulated data did not have them. Although I chose BIC purely pragmatically, the different philosophies for BIC and AICc support the choice. AICc chooses the best fit model, assuming that the 'truth' has many parameters to describe it, while BIC chooses the best true model, assuming that it is of low dimension (Buckland *et al.* 1997). The latter best describes my technique, which searches only for the parameters for behavioural states and partitions, not for the myriad of parameters typically seen in studies such as wildlife habitat use.

Bayesian Information Criteria assumes normally distributed data. However, it may be possible to relax this assumption by using some of the newer model selection procedures, such as WAIC or WBIC (Watanabe 2013).

Behavioural states

The step in the BMS algorithm that grouped the segments into behavioural states used k-means clustering. One might thus suppose that instead of using the iterative searching procedure I used to find new partitions, one could use a clustering technique for the whole method; that is, simply cluster the data from individual locations. Van Moorter *et al.* (2010) used this approach to identify behavioural states of elk, but they did not test it against data generated with known behavioural states.

I tested straight clustering of locations using simulated data, but had limited success. I tested 56 different combinations of clustering methods and distance functions (results not presented) and could not find one that gave useful results. Typically, they either under- or over-estimated the numbers of segments and behavioural states. The few that gave reasonably accurate estimates of the numbers of segments and behavioural states gave wildly incorrect estimates for the actual partition points. Perhaps straight clustering techniques didn't work because they assume that the data set is unordered, whereas my iterative BMS searching technique uses the natural ordering of the data set.

Computational speed

The BMS iterative algorithm I use is a version of direct search optimisation (Kolda *et al.* 2003). One limitation of this algorithm is the slow running time for large n-values. I could not use one of the faster current nonlinear optimisation techniques because they require smooth continuous functions with numerical estimates of first derivatives (Kolda *et al.* 2003). My analysis has numerical noise and discrete parameters (# of states and segments) from which it is not possible to numerically calculate first derivatives. Furthermore, the number of parameters is not known (it is estimated). However, running time for BMS can be minimised by increasing the values of minimum segment length and sampling resolution.

One could also decrease running time by compressing the movement path. For example, if minimum segment lengths of 100 steps for a 10 000-step path are expected, one could compress the path by estimating a mean location for each 10 steps. This would result in a 1000-step path with a minimum segment length of 10. Thus, in the iterative process used to select partition points, one could minimise running time by using a version of the compass search algorithm (Kolda *et al.* 2003) which adds more hierarchical levels. For example, initially sampling every r_1th location, then sampling every r_2th location within the range of two adjacent r_1th locations, and repeating this down to the smallest hierarchical level. However the initial subdivision would still need to place several points within each segment.

Choices of user-set parameters

The first user-set parameter is the minimum segment length, which is a biological choice. Animal have different biological states at various spatial scales. For example, at large spatial scales elk have dispersive and home-range states, while at intermediate scales they switch between exploration, foraging and social bonding behaviours, and at small scales they switch between foraging and searching for new food patches (Fryxell *et al.* 2008). It is likely that at even smaller scales there are different behavioural states, while foraging. The minimum segment length should be set so that the desired scale of states is captured.

The second user-set parameters relate to gaps in data sets. While it may be biologically reasonable to ignore short gaps in the time series, fixed partitions should be used at any gap long enough to contain more than one transition between states. The above-described analysis is then carried out with these partitions fixed in place but releasing the constraint preventing sequential partitions from having the same behavioural. Thus, any behavioural states will be defined consistently across the different path sections.

Similarly, movement paths from multiple individuals can be combined by joining them into one data set, but specifying partitions between each path.

EXAMPLE

I simulated a movement path comprised of four different CRWs (i.e. the behavioural states) distributed through eight segments. The CRWs were each simulated with independent turning angles distributed with a von Mises distribution (Codling *et al.* 2008), and step lengths with a truncated normal distribution with a standard deviation of 0.1 units. Mean step lengths for the four CRWs were 0.94, 1.07, 0.81 and 1.19 units, and mean absolute turning angles were 9.6, 8.5, 12 and 6.7°. The four CRWs were distributed throughout the eight segments in the following order: 1, 2, 3, 1, 4, 1, 2, 3. The length of each segment was 40, 67, 55, 63, 34, 80 and 80 steps. The BMS algorithm was run using a minimum segment length of 20, a sampling resolution of 5, testing # states = 2, 3, 4, 5, 6, 7 and 10. The variables used were speed and absolute value of the turning angle (for brevity, from now on I will refer to 'absolute value of the turning angle' just as 'turning angle').

The results of the BMS analysis agree well with the simulation model. The movement plot shows no obvious segmentation (Fig. 2a), yet the BMS analysis showed that the estimates with the smallest BIC predicted the correct four states and eight segments. The estimated states and segments are close to those of the simulation model generating the path (Fig. 2b, c). The BIC for the estimated parameters was 22 lower than for the expected model, suggesting that the estimated model actually describes the data set better than the model used to generate it. This can be observed in the speed plot (Fig. 2c) in the segments where the estimates and the model are in disagreement.

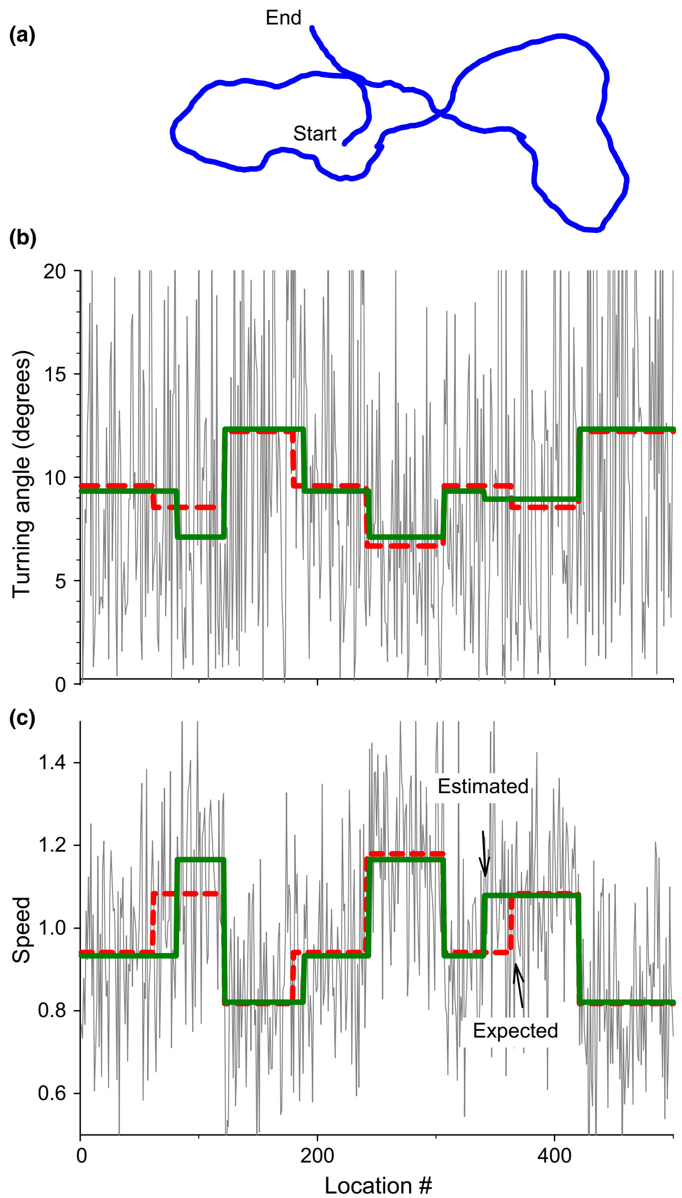


Figure 2 Results from one example simulation: (a) a plot of the movement path, (b) Turning angles and (c) speed. The grey lines show the turning angles and speed for each location, the dashed red lines show the expected values from the model and the solid green lines show the estimated values from the BMS algorithm.

SIMULATION TESTS

Methods

I then tested for the effects of the amount of randomness, the # of states and the # of segments on the performance of the BMS algorithm. I used similar movement models as in the above example. For each choice of the # of states, the model parameters of mean step lengths were distributed equally between a range of 0.6 and 1.4, and mean cosine turning angles were distributed equally between a range of 0.6 and 0.99. For example, for 4 states, mean step lengths for the four different CRWs were 0.6, 0.87, 1.13 and 1.4 units, and mean turning angles were 8, 23, 38 and 53° (not necessarily in the same order as the step lengths). The states were then randomly allocated among the segments, with the restriction that adjacent segments would not have the same states. Segment lengths were uniformly distributed within a range of mean length −15 to mean length +15. For each parameter combination, I modelled 40 paths, each 500 steps long.

I varied parameters as follows. The effects of the amount of randomness were tested by varying the standard deviation of step lengths, using values of 0.01, 0.03, 0.05, 0.1, 0.15, 0.2 and 1. The effects of the # of states were tested by using values of 1, 2, 3, 4, 5, 6, 7 and 8. The effects of the # of segments were tested by using values of 1, 4, 6, 8, 12 and 16. Note that # states = 1 and # segments = 1 is a non-segmented model.

The BMS algorithm was run using a minimum segment length of 20 and a resolution of 5. For each simulation, I estimated the # of states, the # of segments and a measure of model fit. It is not possible to use BIC to compare different simulations because BIC is standardised to the individual data set (Burnham & Anderson 2002). For a measure of model fit, I used the correlation between simulated speed and turning angles and those estimated from the BMS model. This measure is sensitive to estimation of the # of states and the positioning of the boundaries.

RESULTS

I tested a range of simulations using just one CRW and they all gave 1 state and 1 segment, as expected (data not shown).

Increased variation in step length decreases the estimated # of segments and # of states. The BMS algorithm gives a good fit until the coefficient of variation in step length increases past 0.2 (Fig. 3a). Note that at these high levels, the ΔBIC for the expected model was typically about 50. This suggests that the observed model was much more likely to describe the data set than the expected model – i.e. the lack of fit was because the data set was losing information. This is expected because the movement pattern would approach a one-step CRW for high levels of randomness, which is best described with 1 state and 1 segment.

The # of states has a slight effect on the results of the BMS algorithm (Fig. 3b). The fit remains high, but for low # of states, the estimate for # of states and segments is slightly biased upwards. The # of segments also has a slight effect on the results of the BMS algorithm (Fig. 3c). The fit drops slightly at higher # of segments. The estimated # of segments is relatively unaffected, but the # of states is biased upwards at a high # of segments.

Increasing location error (Fig. 3d) decreases estimated numbers of states and segments. Thus, the BMS method degrades gracefully to detecting no segments when the paths are completely random – rather than detecting spurious segments. Thus BMS is robust to error-prone data.

CASE STUDY

The following data set (LaPoint *et al.* 2013) is from a study that analysed corridor use by fisher (*Martes pennanti*). The locations were analysed in that study, but the accelerometer data were not. The movement path is a GPS track of a single fisher released near Albany, New York, USA (Latitude 43° 42.84131, Longitude −73.9027°) on Feb 9, 2010, and retrieved on April 1, 2010. Locations were gathered every 2, 10 or 60 min, depending on the animal's activity (Brown *et al.* 2012). Tri-axial accelerometers measured three-dimensional acceleration in 2.88 s bursts every 3 min. During those bursts, 54 accelerometer readings were taken.

I used the path section from March 12 to 20, estimating acceleration and orientation. Acceleration was estimated by the standard deviation of the vector of the dynamic body acceleration (Qasem *et al.* 2012), over each 2.88 s long burst. Orientation was estimated as the absolute value of the angle between vertical and the orientation of the collar, with orientation being averaged over each 2.88 s long burst. Thus the orientation angle combines pitch and roll. Vertical was defined as the mean orientation while the animal was travelling faster than 30 m/min between successive locations. Mean orientation values were not sensitive to the exact definition of vertical since defining vertical as the mean orientation while the animal was travelling slowly had little effect on the mean values.

Estimated locations were added to fill in gaps where the animal was stationary to ensure that all time periods are weighted equally. The estimated stationary locations were normally distributed with a variance as calculated by the GPS collar. The resulting path had 1400 locations taken approximately every 3 min, with three large gaps due to missing locations. Thus, there were four separate sections in the path.

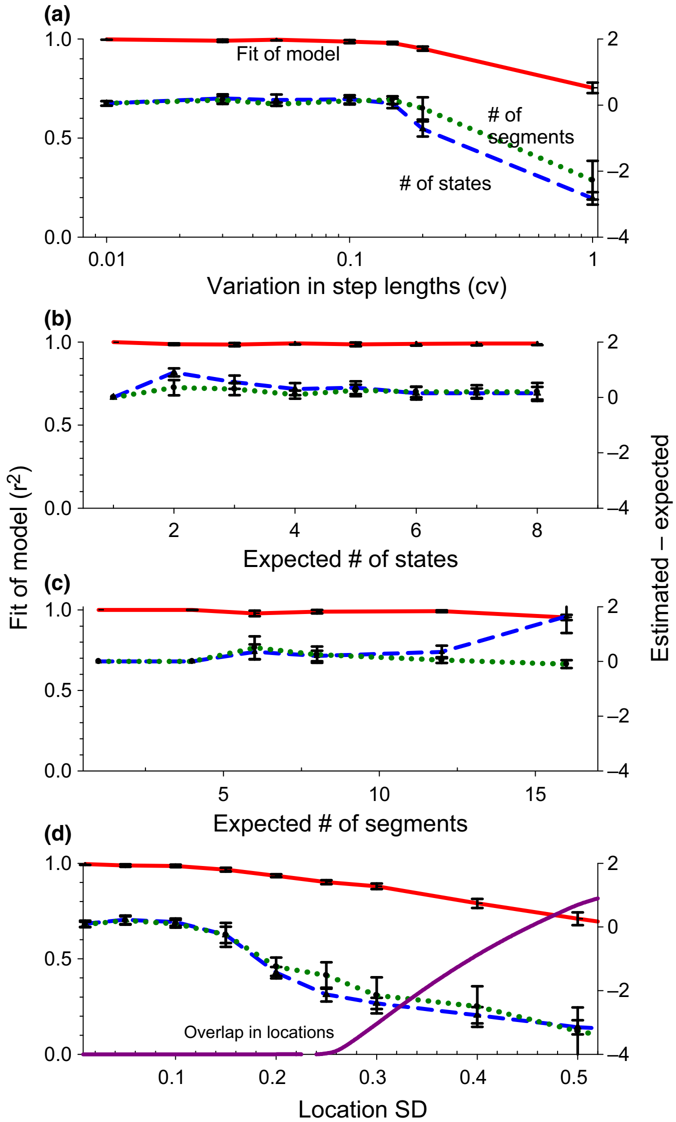


Figure 3 Results from simulations comparing the effects of (a) randomness, (b) # of states, (c) # of segments and (d) location error. In (d) the purple line shows the proportion of overlap in location estimates due to location error. This uses the left axis. Thus, at a location SD of 0.5 there is an almost total overlap (0.85) in 95% per cent error ellipses between adjacent locations.

I used four variables: the turning angle, speed, acceleration and orientation. These were estimated between the current and previous location. Speed and orientation were log-transformed, and the resulting means and confidence intervals were back-transformed for display. The BMS procedure was run using a minimum segment length of 15 and a sampling resolution of 4.

Employing only the movement path statistics (I will call this segmentation A), the most likely model contained 4 states divided into 28 segments (Fig. 4a). State 1 represented fast and straight movement (high speed and low turning angles),

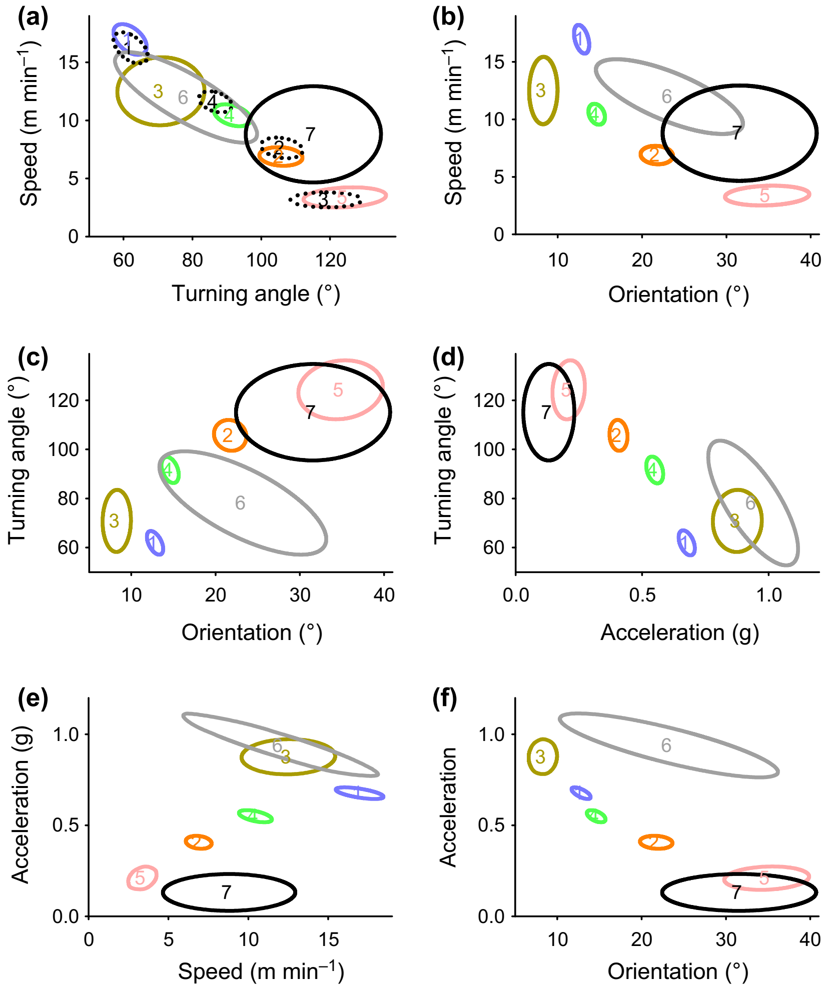


Figure 4 Comparisons of the mean values for the two segmentation types for all six combinations of variables. The ellipses are 95% confidence ellipses of the means for each behavioural state. The dotted ellipses with black numbers (a) represent the segmentation using only movement variables and the solid ellipses the segmentation using both the movement variables and the accelerometer data. The colouring and numbering of the behavioural states are the same (except for black) as in Figs 5 and 6.

state 3 represented the animal being stationary (slow speed and high turning angles), and the other two states were in-between. When the accelerometer data were added (segmentation B), the most likely model contained 7 states divided into 41 segments (Figs 4b, 5 and 6).

The addition of the extra variables gave results showing extra complexity in the organisation of behavioural states. For example, when only speed and turning angles are considered, most of the states fall into a line. This suggests that there is a one-dimensional continuum of behaviours from fast and straight to slow and tortuous (Fig. 4a). However, when both speed and orientation are considered, the states fall into two lines. This suggests at least 2 dimensions differentiating behavioural states (Fig. 4b).

DISCUSSION

Combining different types of data

Combining location data with other ancillary data in one analysis allows us to detect/partition behaviours in ways previously not possible. Using movement data allows one to study behaviour at larger scales and in more varied habitats than direct observations allow. However, when biologists actually observe animals, they typically see more types of behaviours than they detect by analysing movement data. Ancillary sensor data can quantify these kinds of behaviours.

For example, Jiang & Hudson (1993) observed elk feeding in two habitat types and detected 10 different behaviour states. In contrast, Fryxell *et al.* (2008) analysed elk movement data from all habitat types (even while animals moved 5–7 km per day) but detected only two types of movements. Shannon *et al.* (2008) observed African elephants (*Loxodonta africana*) during the day and only when animals were visible, then classified their behaviours into five categories. In contrast, Roever *et al.* (2014) gathered data every hour all day and night for 3 years over an area of 300 km^2, then used state space modelling to partition elephant telemetry locations into only two types of movements. Such different approaches yield vastly different results.

One might wonder whether using inappropriate ancillary data might create spurious segmentation. Adding data vari-

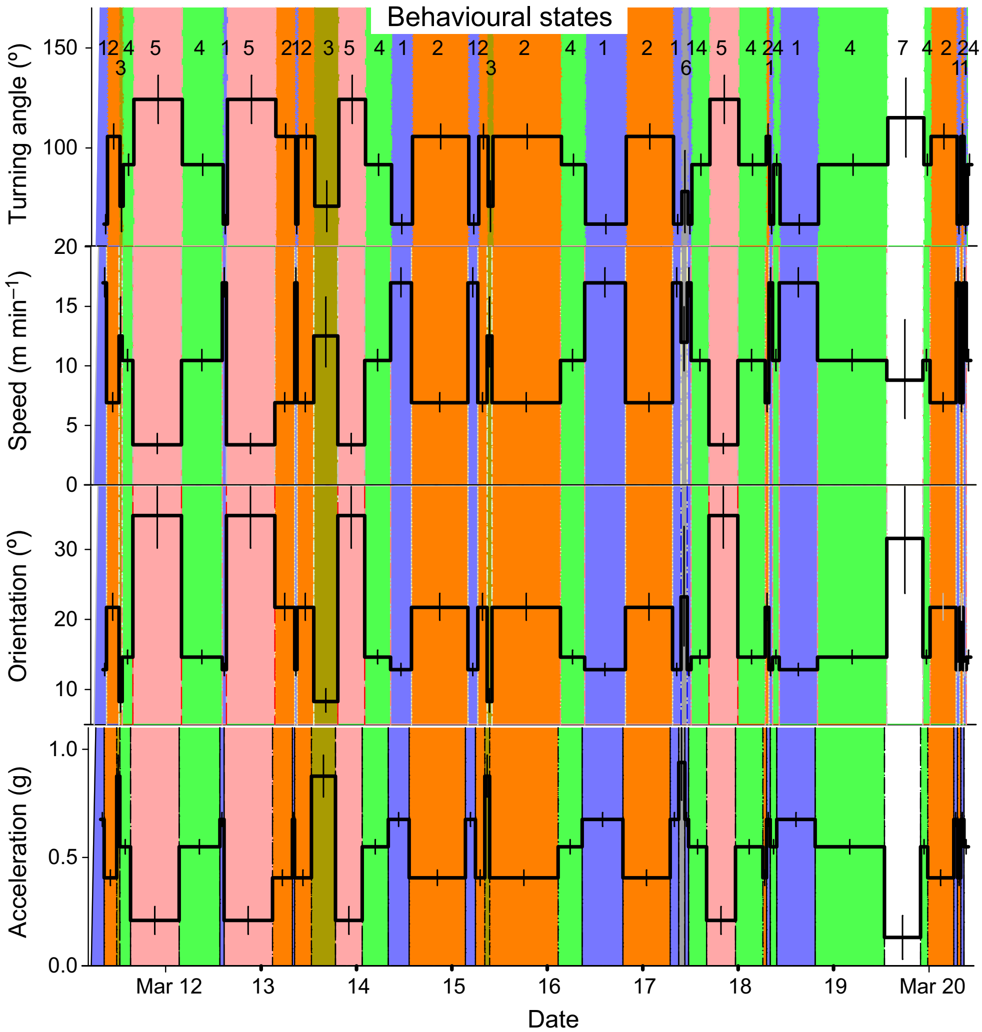


Figure 5 Summary statistics for segmentation of a fisher movement path, plotted as a function of time. The variables used were movement path statistics and accelerometer data. Data are (a) mean turning angles, (b) speeds, (c) acceleration and (d) standard deviation of acceleration, for each of the 6 states, divided into 34 segments. The colouring and the numbers at the top are the different behavioural states, and are the same (except for white) as used in Figs 4 and 6. The bars are 95% confidence intervals.

ables that do not reflect behavioural states is equivalent to adding noise to the data set. However, I showed that adding errors causes BMS to discover fewer, not more segments.

Combining behavioural with location data in one analysis also allows us to combine information taken at different scales. Current-technology GPS collars can locate an animal to at best 5 m, but more typically 30 m. Thus only gross changes in behaviour can be detectable from such data. On the other hand, behavioural sensors can measure variation in movement at very small spatial and temporal scales. For example, accelerometers can sample acceleration at rates of up to 30 time per second (Wilson *et al.* 2006), and the collar only needs to shift slightly in position to register a change. This allows biologists to determine more behavioural states. For example, Nathan *et al.* (2012) used tri-axial acceleration data to classify seven visually confirmed behavioural states in free-flying griffon vultures (*Gyps fulvus*). Thus, use of my statistical technique, which can combine location and behavioural data, will allow biologists to classify movements in ways that incorporate both changes in location as well as orientation.

Habitat use

BMS can be used to resolve one of the key issues for conservation biology of rare species – identifying important habitats. Typically, habitats are evaluated using Resource Selection Functions, which involve predictive modelling of locations based on the habitats (Boyce *et al.* 2002). However, animal habitat use also depends on what behavioural states the animals are in, because it is a combination of internal and external factors that determines where animals live (Nathan *et al.* 2008). For example Roever *et al.* (2014) showed that spatial predictions of habitat use by African elephants were significantly enhanced when movements were first partitioned into two behavioural states (encamped vs. exploratory), after

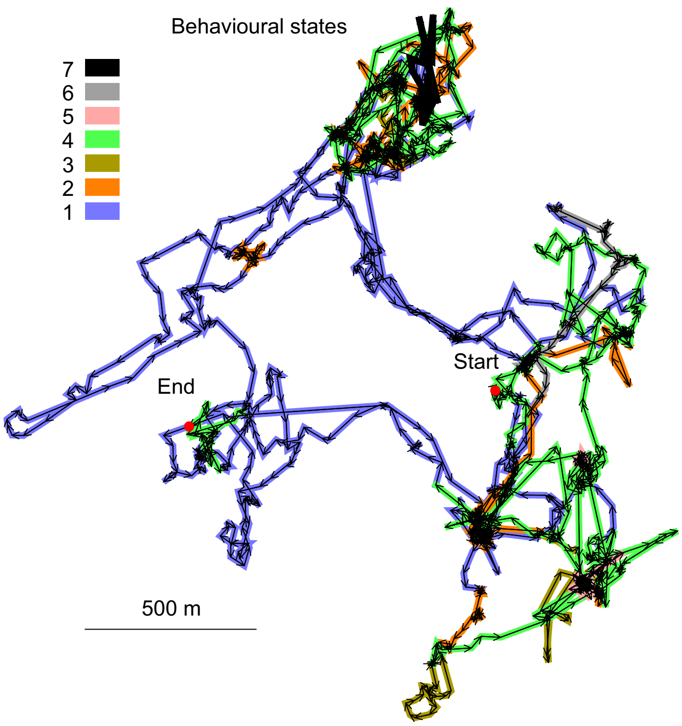


Figure 6 The movement path of the fisher showing the segmentation. The colouring (except for black) and numbering of the behavioural states are the same as used in Figs 4 and 5.

which habitat use predictions were made for each of those behavioural states. The capability to combine behaviour and habitat use can be improved by using BMS to first classify behaviours using movement and ancillary sensor data before modelling habitat use.

Applications

The underlying BMS method analyses data that occurs in a time series. I applied it to turning angles and speeds calculated from a animal movement pattern, but the BMS method can be applied to any kind of sequential behavioural observations, completely independently from animal movements. A large literature exists regarding how to split animal behaviour into bouts when observations are times that individual behaviours occur. For example, Schaefer & Messier (1995) observed pawing acts by muskoxen and divided those observations into two states – bouts of pawing separated by gaps. They used assumptions about the distribution of times between events to separate the events into discrete categories (Sibley *et al.* 1990). Since their method requires strong assumptions about underlying distributions, it is statistically powerful. However, its limitation is that it does not apply when times between bouts are not exponentially distributed. One could create a more robust analysis by applying BMS, using the variable of time between events, and could also incorporate other behaviour measurements into the analysis.

The BMS algorithm detects behavioural states at one spatial scale; however animals show variation at different spatial scales. It is possible to incorporate BMS in a hierarchical analysis. One approach would be to detect the biological domains using other techniques (Nams 2005), and then to apply BMS at each spatial scale. Another would be to use BMS to directly detect domains by applying it hierarchically. This would involve first applying BMS to the whole movement path to detect partitioning, and then applying it to each behavioural state (combining analyses for all segments of that state) to detect partitioning within that state. This hierarchical application could be repeated for lower and lower levels of behavioural state partitioning.

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